Anyu Pan

pananyu@engineering.upenn.edu · anyupay.github.io · +1 917-754-3292

Research Interests

Agentic AI · AI for Healthcare · Computer Vision · Computational Drug Discovery

Education

University of Pennsylvania 2024 – 2027
M.S. in Computer and Information Science · *M.S. in Bioengineering* *USA*

Northeastern University 2020 – 2024
B.Eng. in Biomedical Engineering *China*
Joint Undergraduate Programme with University of Dundee, UK · First Class Honours

Research Experience

Shen Lab, University of Pennsylvania Jun 2026 – Present
Graduate Student Researcher, Penn Engineering (CIS) & Penn Medicine · *PI: Li Shen, Ph.D.*

Alzheimer's Disease Drug Discovery Jul 2026 – Present

- Built an **AGORA-centered target database** integrating BindingDB, ChEMBL, UniProt, RCSB PDB, Drugs@FDA, and ClinicalTrials.gov using **DuckDB**; linked **967 targets, 452,249 compounds, and 1,430,260 measurements**
- Curated a PIK3CA potency dataset and fine-tuned the **Boltz-2 affinity head**; improved Pearson r from **0.248 to 0.560** and reduced MAE from **0.900 to 0.706** on **469 held-out compounds**
- Developed and deployed AGORA Target & Drug Evidence Explorer on **Azure**, providing a public platform for AD target assessment, drug-target evidence queries, curated data exports, and access to fine-tuned models and evaluation results

Mouse Scratching Classification with Video Transformers Jun 2026 – Present

- Fine-tuned **Video Swin Transformer** and adapted **InternVideo-Next** with low-rank adaptation (LoRA) for **Scratch-AID+**, a framework for automated mouse scratching detection
- Evaluated InternVideo-Next on test videos from four external labs; achieved an average F1 score of **81.19%** for scratching detection, outperforming the Scratch-AID baseline by **14.88 percentage points**

Advancing Glioma Classification with Swin Transformers Sep 2023 – May 2024
Honours Project, University of Dundee & University of Cambridge

- Integrated **Swin Transformer blocks** into a WSI-based glioma classification framework, improving accuracy to 78.03% (from 73.11% with a standard ViT)
- Systematically evaluated multiple **optimizers** (SGD, Momentum, AdaGrad, RMSprop, Adam, and Lion) to identify optimal performance-efficiency trade-offs
- Reduced **GPU memory usage by >60%**, enabling larger batch sizes and more efficient GPU training

Publications

[1] Yang, S.*, **Pan, A.***, Shin, H.*, Chen, H.*, Zhang, Z., Shad, A., Huang, W., Liu, Y., Liu, W., Luo, H., Xu, C., Chen, J., Xu, M., Luo, W., Yu, H., & Shen, L. (2026). Towards Robust Automated Mouse Scratching Behavior Detection for Itch Research. *AI4BioSig 2026* (*IEEE BIBM 2026 Workshop*), accepted. [Paper]

[2] **Pan, A.**, Zhai, T., Lee, J., Zheng, C., Kim, D., & Shen, L. (2026). Integrating Multisource Target Evidence with Machine-Learning Potency Prediction for Alzheimer's Disease Drug Discovery. Submitted to *AMIA 2027 Amplify Informatics Summit* (regular paper). [Paper]

[3] Qiu, Z., **Pan, A.**, & Qi, L. (2023). Incorporating dynamic attention gating mechanism and pre-trained embedding for Chinese clinical named entity recognition. *Procedia Computer Science*, 226, 113–119.

* Equal contribution (co-first authors).

Selected Research & AI Projects

WeatherWise Trip Concierge · *Google/Kaggle AI Agents Capstone* Jun 2026 – Jul 2026

- Built a **Google ADK travel-planning agent** with **MCP tools** to retrieve forecasts, score daily weather risks, and recommend nearby cities
- Implemented **rule-based risk scoring** and Pydantic input validation; tested prompt-injection and missing-forecast cases
- Created **AGENTS.md**, project context documentation, and reusable **agent skills**; deployed the web app on **Google Cloud Run** using Docker

Decoding Finger Flexion from ECoG · *Brain-Computer Interface* · *Member* Mar 2025 – May 2025

- Developed a **convolutional encoder–decoder** to predict five-finger flexion from ECoG (BCI Competition IV dataset)
- Built signal preprocessing with bandpass/notch filters, **Morlet wavelets**, downsampling, and lag correction
- Achieved mean Pearson correlation of 0.624 on the held-out test set using a combined **MSE + cosine loss**

Biomedical Image Analysis · *Member* Nov 2024 – Dec 2024

- Preprocessed MRI scans and tuned **N4 bias field correction** to improve segmentation inputs
- Quality-checked 3D training patches, labeled MRI in **ITK-SNAP**, and merged team labels into ground-truth masks
- Trained and evaluated an **nnU-Net** segmentation pipeline, refining hyperparameters to improve accuracy

8th National College Student Biomedical Engineering Innovation Design Competition · *Core Member* May 2023 – Jun 2023

- Integrated **YOLOv5 and Transformer models** for rapid colorectal polyp detection and diagnosis
- Reduced the missed detection rate from **22% to 1%** in simulations

Selected CS/ML Projects

Distributed Search Engine · *Internet and Web Systems* · *Member* Nov 2025 – Dec 2025

- Hardened a crawler with HTTP/socket timeouts and robots handling; crawled **100K+ pages in ~9 hours**
- Optimized distributed **PageRank** using per-partition KVS writes; fixed persistence and Flame partitioning, reducing client churn
- Added autocomplete and normalized URLs, titles, and snippets; deployed on **AWS EC2** and benchmarked instance types

Spotify Popularity Prediction · *Big Data Analytics* · *Member* Oct 2024 – Dec 2024

- Preprocessed **114,000 Spotify records** and used Plotly for interactive exploration of popularity-related features
- Trained and tuned **Random Forest and XGBoost** regression models for song popularity prediction

Knock2Know: Audio Classification · *Applied Machine Learning* · *Member* Sep 2024 – Dec 2024

- Collected and preprocessed audio across recording devices, times of day, and knocking techniques to improve robustness
- Extracted **OpenAI Whisper** features and trained a shallow CNN with augmentation and early stopping; achieved **0.71 accuracy** on unseen samples

16th Computer Design Competition for Chinese College Students · *Member* Apr 2023 – Aug 2023

- Developed a **Flutter** game about traditional Chinese medicine; led debugging, documentation, and regression testing

Breaking the Wordle · *Mathematical Contest in Modeling* · *Core Member* Feb 2023

- Modeled Wordle trends with **Prophet** and built predictive models using multi-objective regression and **LASSO**.

Teaching Experience

Teaching Assistant, Developmental Engineering of Tissues · *Penn Engineering* Aug 2024 – Dec 2024

- Graded code and reports for a neural crest migration model, covering Voronoi adjacency, border-cell detection, force balances, and numerical integration; provided detailed feedback
- Held weekly office hours on debugging and model design; graded quizzes to assess preparation and progress

Awards & Honors

ASA Data Insights Award (top 0.1%), Mathematical Contest in Modeling, International Nov 2023

First Prize (top 4%), 16th Computer Design Competition for Chinese College Students, National Aug 2023

Outstanding Winner (top 0.2%), **COMAP Scholarship Award & ASA Award**, Mathematical Contest in Modeling, International May 2023

Skills

Programming	Python, Java, C/C++, SQL, R, MATLAB, HTML, JavaScript, CSS, Dart
Machine Learning	PyTorch, scikit-learn, XGBoost, ViT, Video Swin Transformer, InternVideo-Next, LoRA, Boltz-2, YOLO, nnU-Net, NumPy, SciPy, pandas
Agent Development	Google ADK, MCP, Agent Skills, Docker, Google Cloud Run
Research Infrastructure	RDKit, DuckDB, Azure, Slurm
Other	AWS, Flutter, Qt, Git, LaTeX, Arduino IDE, InVesalius, ITK-SNAP, SolidWorks, Proteus

Academic Exchange

Osaka University May 2021 – Aug 2021
Exchange Program in Biotechnology *Japan*